

LECTURE 2

Compressibility, Vapor Pressure and Surface Tension

Reference: Munson et al., Fundamentals of Fluid Mechanics, Chapter 1
Wladimir E. Banda-Barragán

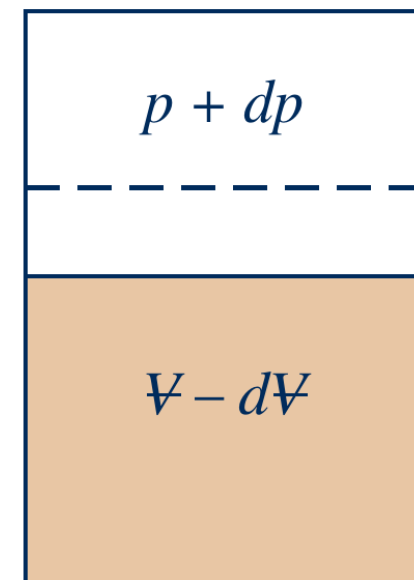
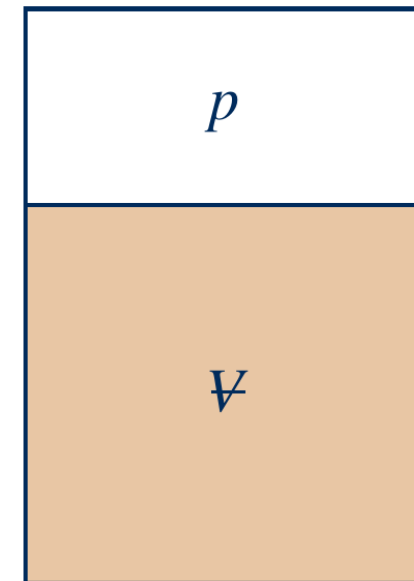
Compressibility of fluids

How easily can the volume, and therefore the density, of a given mass of fluid be changed when the pressure changes?

The property that answers this is the **bulk modulus**:

$$E_v = - \frac{dp}{dV/V}$$

- dp is the pressure change that produces the change in volume dV of a volume V .
- The negative sign is there because an increase in pressure reduces the volume.



The bulk modulus

A given mass has $m = \rho V$, so a decrease in volume is an increase in density, and the definition can be written in terms of the density:

$$E_v = \frac{dp}{d\rho/\rho}$$

$$[E_v] \doteq FL^{-2} : \text{Pa}$$

- **A large bulk modulus means the fluid is nearly incompressible:** a large pressure change is needed to produce a small change in volume.
- The bulk modulus is also called the **bulk modulus of elasticity**.

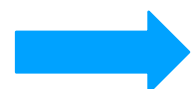
How incompressible is a liquid?

Example: Water at atmospheric pressure and 15.6 °C:

$$E_v = 2.15 \text{ GPa}$$

$$\Delta p = 21.5 \text{ MPa} \Rightarrow \frac{\Delta V}{V} = 1\%$$

- It takes 21.5 MPa, more than 200 atmospheres, to compress a unit volume of water by 1%.
- Since such pressures are rare in practice, **liquids can be treated as incompressible in most engineering applications.** The bulk modulus increases as a liquid is compressed, but the value near atmospheric pressure is the one of interest.

 **Liquids: incompressible for most purposes**

Compression and expansion of gases

For a gas, the relation between pressure and density depends on the process.

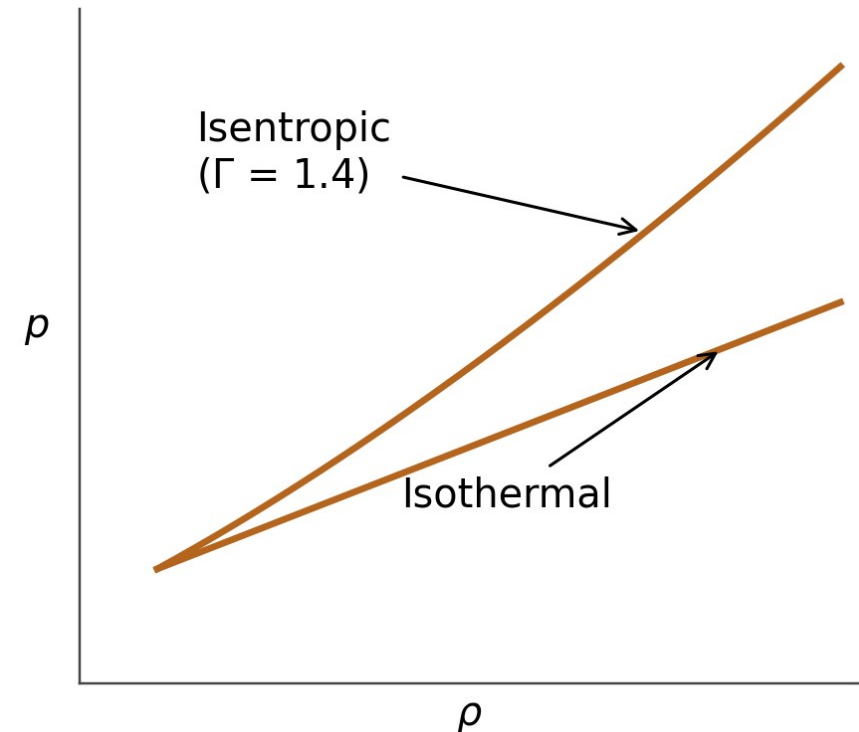
At constant temperature, **an isothermal process**:

$$\frac{p}{\rho} = \text{constant}$$

Frictionless and with no heat exchanged, **an isentropic process**:

$$\frac{p}{\rho^\Gamma} = \text{constant}$$

$$\Gamma = \frac{c_p}{c_v}, \quad R = c_p - c_v$$



Pressure-density variation for the two processes.

Γ is the ratio of the specific heats, 1.40 for air. (The textbook writes it k , but I don't like that notation)

As in the ideal gas law, p must be absolute.

The bulk modulus of a gas

Differentiating each relation and substituting dp/dp into the definition of the bulk modulus:

$$E_v = p$$

isothermal process

$$E_v = \Gamma p$$

isentropic process

In both cases the bulk modulus varies directly with pressure. For air at standard atmospheric conditions, with $p = 101.3 \text{ kPa (abs)}$ and $\Gamma = 1.40$:

$$E_v = (1.40)(101.3 \text{ kPa}) = 142 \text{ kPa}$$

 **Air is about 15,000 times as compressible as water**

Incompressibility — demonstration



Bulk Modulus: Although all fluids are compressible, some fluids are relatively difficult to compress.

This is particularly true of liquids.

When a water-filled balloon is squeezed, the volume of the water within the balloon remains essentially the same.

This is true because the bulk modulus of water is very large ($2.15 \times 10^9 \text{ N/m}^2$) and the pressure applied to the balloon is relatively small.

Hence, although the squeezed balloon may take on a variety of different shapes, the volume of the water within it is virtually unchanged.

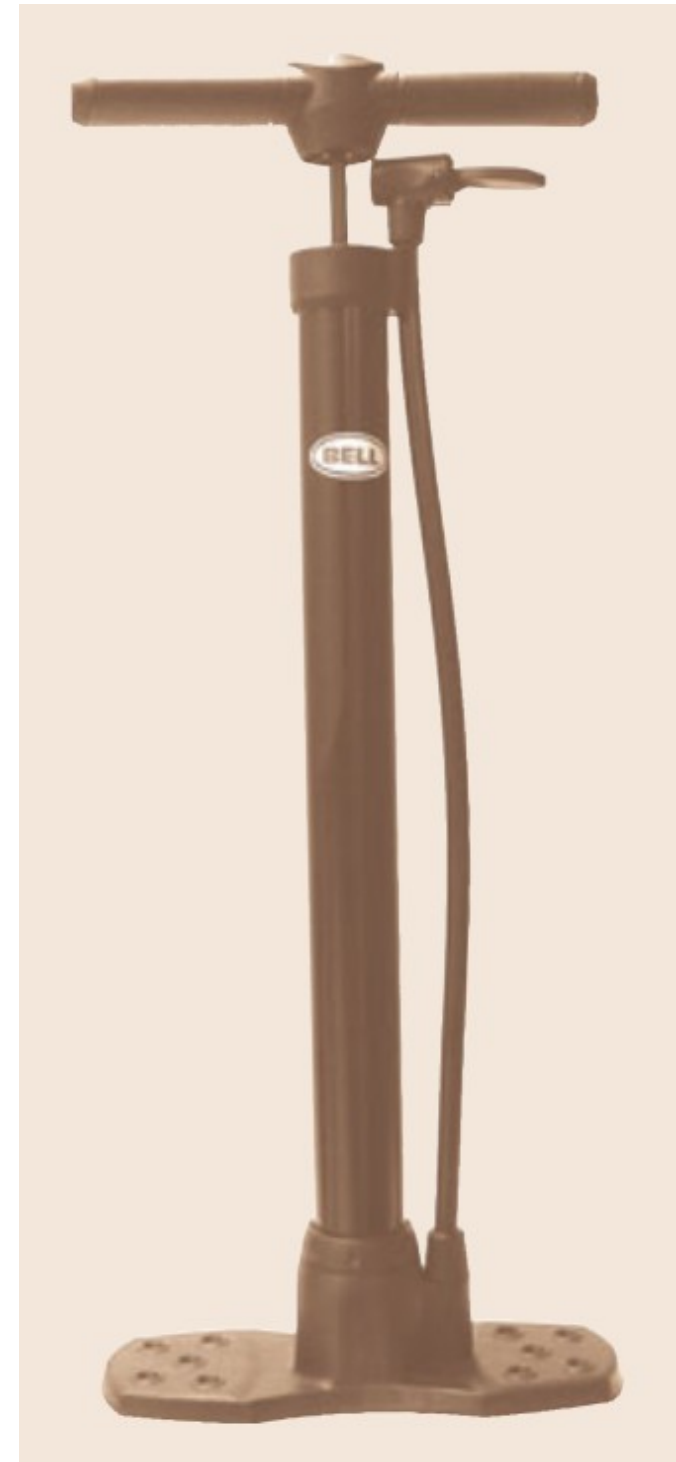
Example — isentropic compression of a gas

A cubic meter of air at an absolute pressure of 101.3 kPa is compressed to half that volume by a tire pump. Determine the final pressure, assuming the compression is isentropic. For air $\Gamma = 1.40$.

$$\frac{p_i}{\rho_i^\Gamma} = \frac{p_f}{\rho_f^\Gamma} \quad \Rightarrow \quad p_f = \left(\frac{\rho_f}{\rho_i} \right)^\Gamma p_i$$

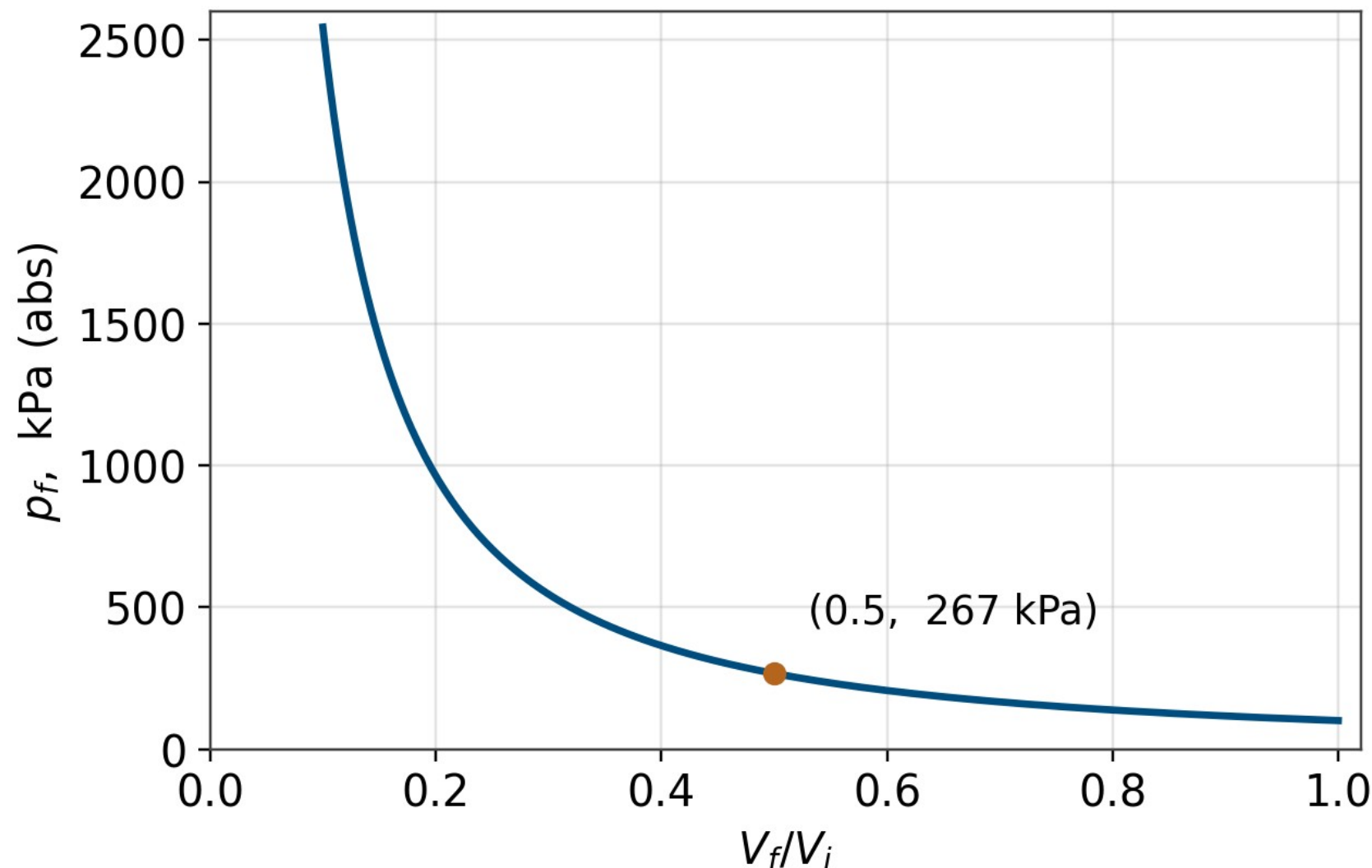
Halving the volume doubles the density, since the mass $m = \rho V$ is constant. With $\Gamma = 1.40$ for air:

$$p_f = (2)^{1.40} (101.3 \text{ kPa}) = 267 \text{ kPa (abs)}$$



Example — isentropic compression, continued

Repeating the calculation over a range of volume ratios gives the curve below.



- Air is easy to compress compared with a liquid, but it still takes considerable pressure to reduce its volume appreciably:
- In a car engine the compression ratio is about $V_f / V_i = 1/8 = 0.125$, which needs about 1.9 MPa.

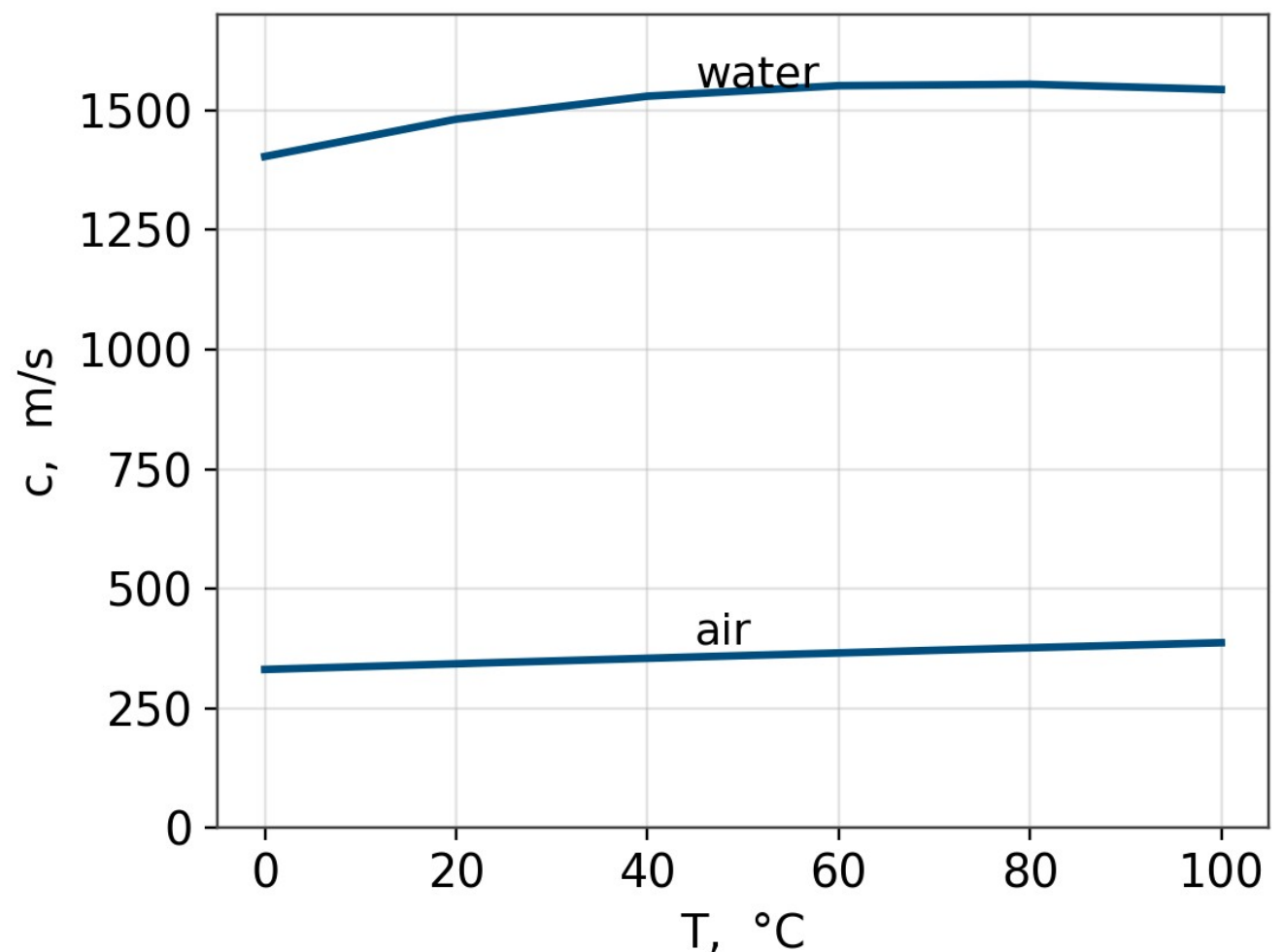
Speed of sound

- Because fluids are compressible, disturbances travel at a finite speed.
- Closing a valve at the end of a pipe is not felt instantly upstream, and the pressure change made by a loudspeaker takes time to reach the listener.
- That speed is the acoustic velocity, or speed of sound.

$$c = \sqrt{\frac{dp}{d\rho}}$$

or, in terms of the bulk modulus:

$$c = \sqrt{\frac{E_v}{\rho}}$$



Speed of sound in water and in air.

Speed of sound in gases

The disturbance is small, so there is negligible heat transfer and the process is isentropic, with $E_v = \Gamma p$. Using the ideal gas law:

$$c = \sqrt{\frac{\Gamma p}{\rho}} = \sqrt{\Gamma R T}$$

For an ideal gas, the speed of sound depends only on the absolute temperature.

$$c_{\text{air}} = 343 \text{ m/s} \quad (20^\circ\text{C})$$

$$c_{\text{water}} = 1481 \text{ m/s} \quad (20^\circ\text{C})$$

Sound travels much faster in water than in air.

If a fluid were truly incompressible, E_v would be infinite and so would the speed of

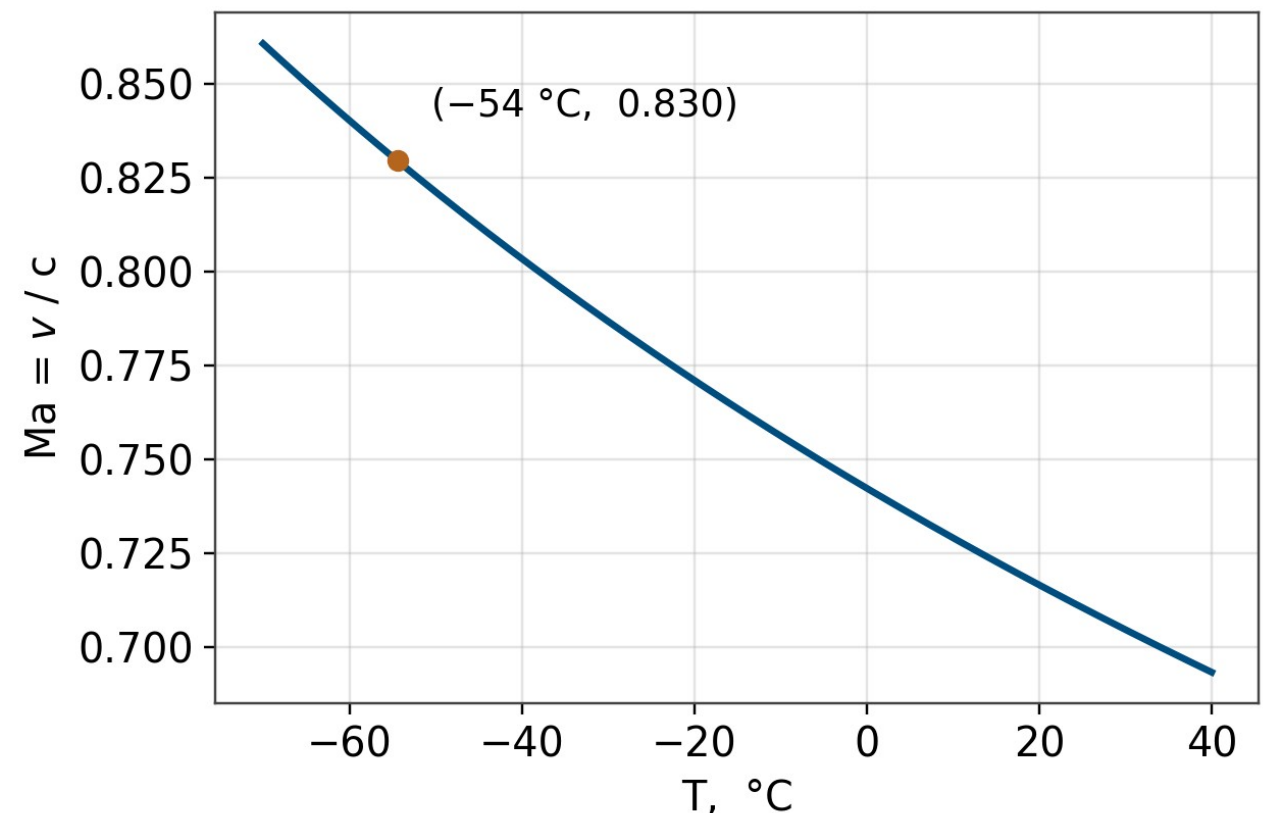
Example — speed of sound and the Mach number

A jet aircraft flies at 885 km/h at an altitude of 10.7 km, where the temperature is $-54\text{ }^{\circ}\text{C}$, that is 219 K. Determine the ratio of the speed of the aircraft to the speed of sound at that altitude. For air take $\Gamma = 1.40$ and $R = 286.9\text{ J}/(\text{kg}\cdot\text{K})$.

$$c = \sqrt{(1.40)(286.9)(219)} = 296\text{ m/s}$$

$$v = \frac{885\text{ km/h}}{3.6} = 246\text{ m/s}$$

$$\text{Ma} = \frac{v}{c} = \frac{246}{296} = 0.83$$



➡ **Ma < 1 subsonic · Ma > 1 supersonic**

As fast as a speeding bullet



Video courtesy of Vision Research, Inc.

- The artillery shell shown is **supersonic**, and the **shock wave it creates shows it**: a thin line ahead of the shell, bent backwards as it wraps around the top and bottom into a blunt wedge.
- Across the shock wave the fluid properties — pressure, temperature, density — change considerably.
- The sonic boom of supersonic flight is that pressure difference across the shock wave, as it goes past the observer.
- The character of the flow past an object depends strongly on whether it moves slower than the speed of sound (subsonic) or faster than it (supersonic).

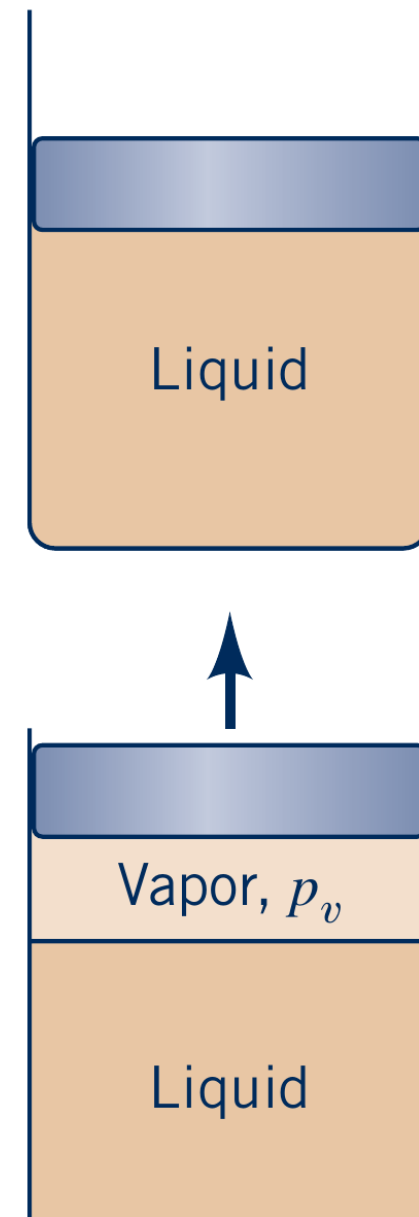
Vapor pressure

Some molecules at the surface of a liquid have enough momentum to overcome the cohesive forces and escape.

In a closed space above the liquid a pressure develops as they do.

At equilibrium, as many molecules return as leave: the vapor is saturated, and the pressure it exerts on the surface is the vapor pressure, p_v .

Vapor pressure is closely tied to molecular activity, so it depends on temperature.



Boiling

Boiling is the formation of vapor bubbles within the fluid. **It begins when the absolute pressure in the liquid reaches the vapor pressure.**

$$p = p_v \implies \text{boiling}$$

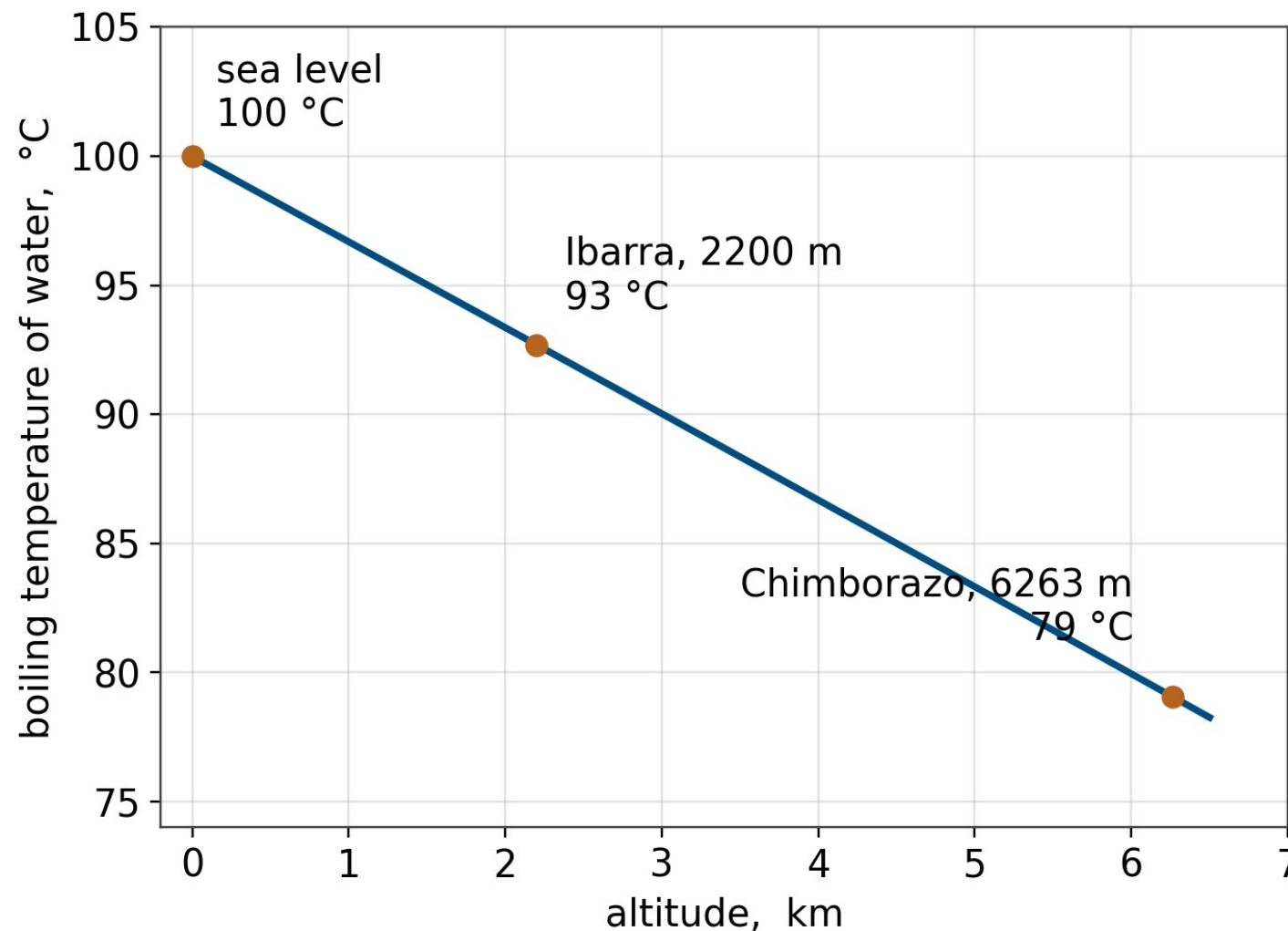
Boiling can therefore be produced in two ways: by raising the temperature at a given pressure, or by lowering the pressure at a given temperature. The second is what matters in a flowing liquid.

Atmospheric pressure falls with altitude, so water boils at a lower temperature the higher we are — a familiar fact in this country:

Guayaquil: $z = 0 : p_{\text{atm}} = 101.3 \text{ kPa} \Rightarrow T_{\text{boil}} = 100^\circ\text{C}$

Ibarra: $z = 2200 \text{ m} : p_{\text{atm}} = 77.5 \text{ kPa} \Rightarrow T_{\text{boil}} = 93^\circ\text{C}$

Boiling temperature in Ecuador



Ibarra sits at 2200 m: the atmospheric pressure is 77.5 kPa and water boils at 93 °C.

On the summit of Chimborazo, 6263 m, the pressure is 45.5 kPa and water boils at 79 °C.

The curve combines the standard atmosphere with the vapor-pressure curve of water.

$$z = 2200 \text{ m} : p_{\text{atm}} = 77.5 \text{ kPa} \Rightarrow T_{\text{boil}} = 93^\circ\text{C}$$

$$z = 6263 \text{ m} : p_{\text{atm}} = 45.5 \text{ kPa} \Rightarrow T_{\text{boil}} = 79^\circ\text{C}$$

Cavitation

In a flowing liquid **the pressure can fall locally to very low values:** through the narrow passages of a valve, in a pump, or over a propeller blade.

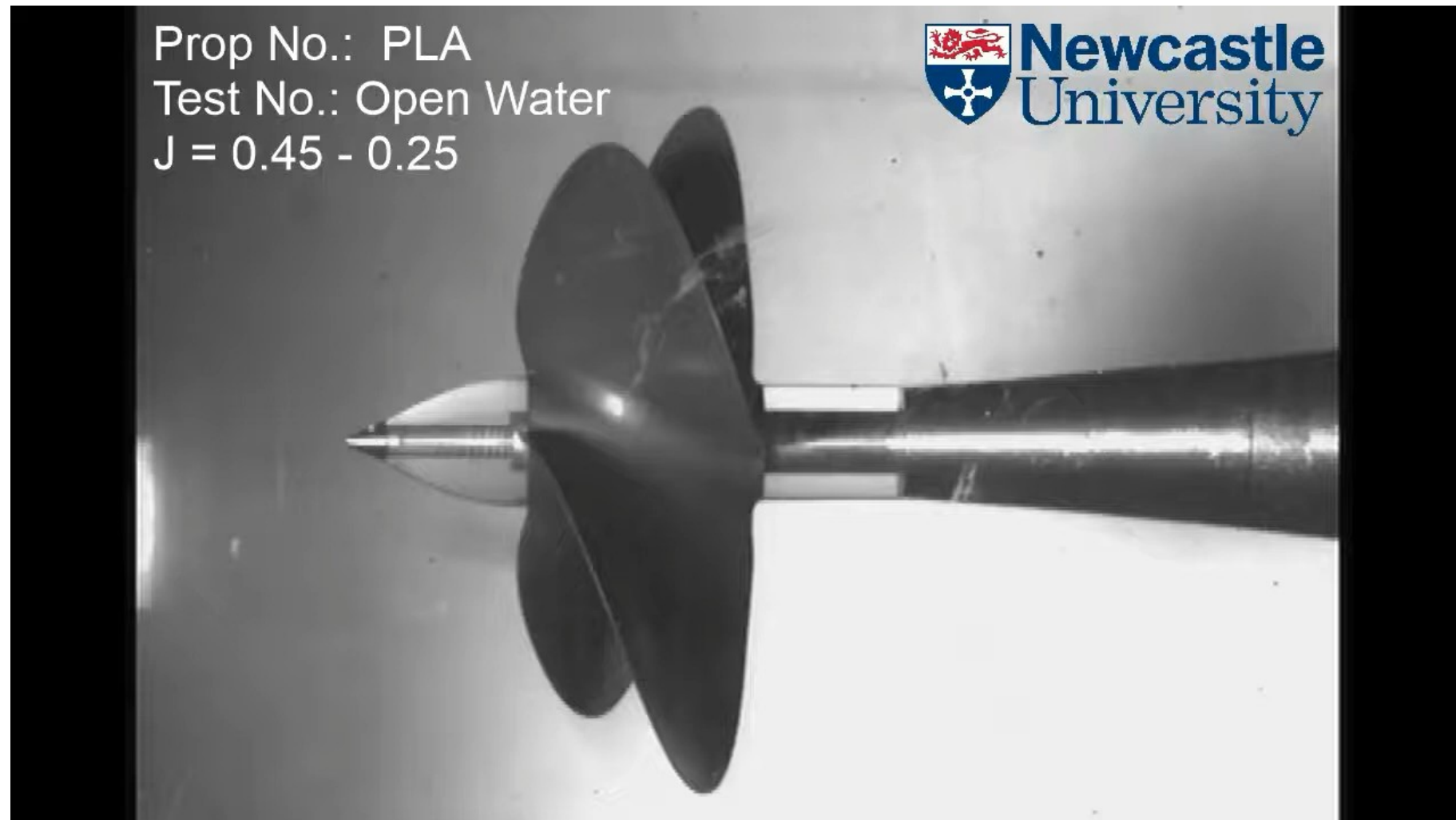
If it reaches the vapor pressure, the liquid boils.

- Vapor bubbles form in the low-pressure region
- They are swept into regions of higher pressure
- There they collapse, with enough intensity to cause structural damage

This is cavitation. It erodes pump impellers and propellers, and **it is one reason for keeping the pressure above the vapor pressure everywhere in a design.**

Cavitation — demonstration

Credits: <https://www.youtube.com/watch?v=jEFELoyuap8>



- When the local pressure in a flowing liquid falls to the vapor pressure, the liquid boils and vapor bubbles form.
- Carried into a region of higher pressure, the bubbles collapse violently.
- The collapse produces very high local pressures, which erode propellers, pump impellers and valve seats.

Surface tension

At the interface between a liquid and a gas, or between two immiscible liquids, forces develop that make the surface behave as if it were a stretched membrane.

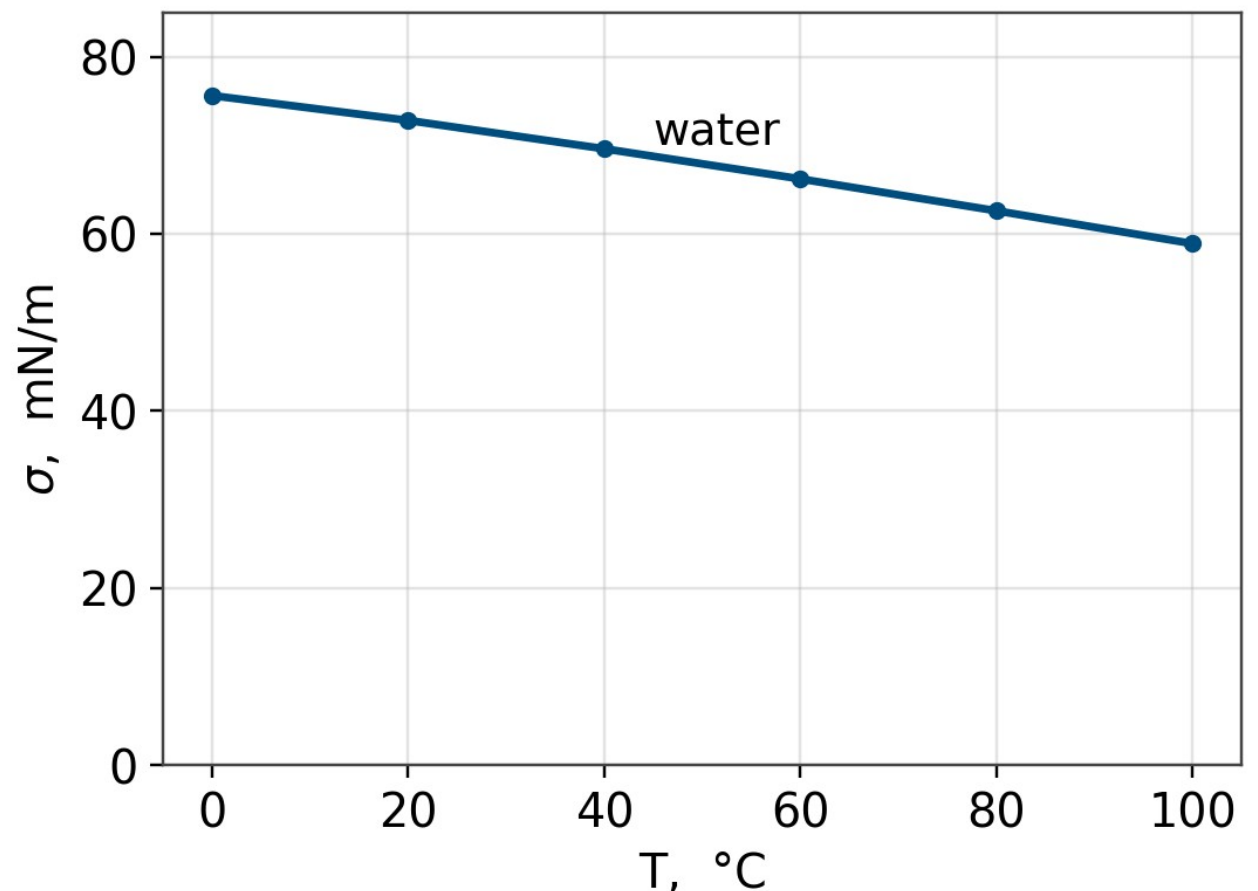
Molecules inside the liquid are pulled equally in all directions.

Molecules at the surface are not: the net force is towards the interior.

The intensity of the molecular attraction per unit length along any line in the surface is the surface tension, σ .

$$[\sigma] \doteq FL^{-1} : \text{N/m}$$

For a given liquid, σ depends on temperature and on the fluid on the other side of the interface, and it decreases as the temperature increases.



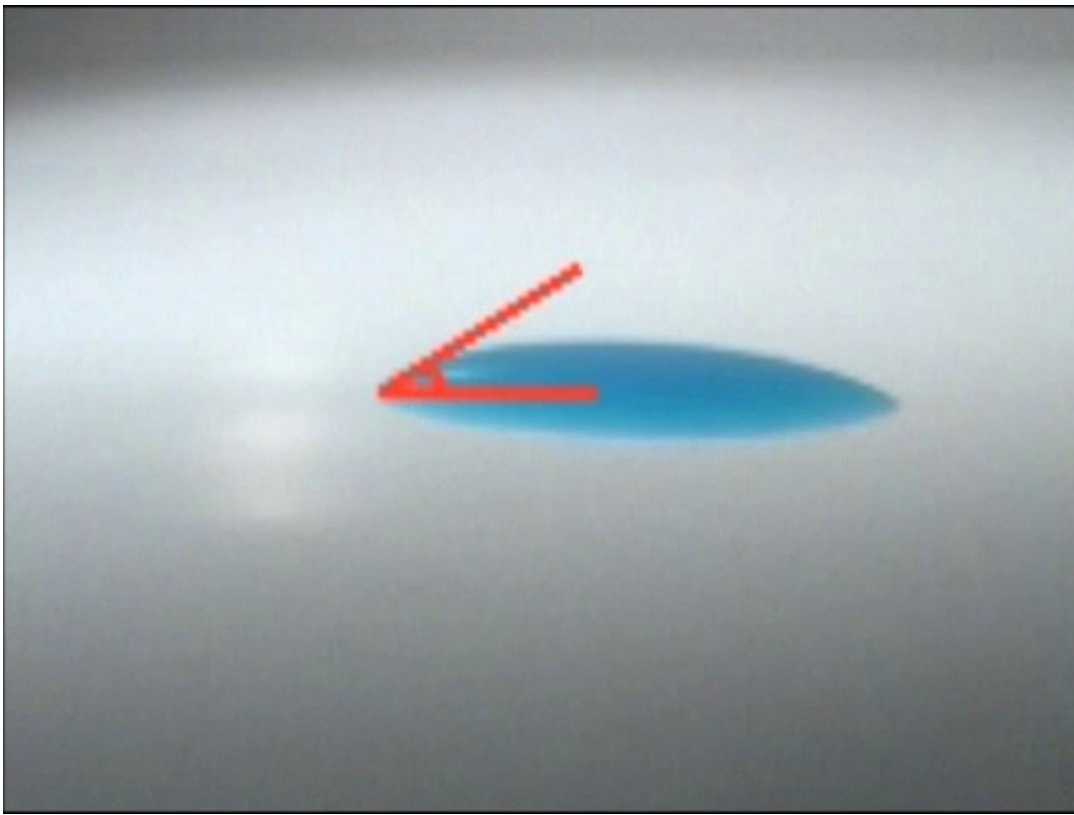
Surface tension — floating razor blade



Floating razor blade.

- A razor blade is denser than water, and yet it can be made to rest on the surface.
- It is not floating in the sense of Archimedes: **it is supported by the surface tension of the water.**
- The surface behaves as though it were a stretched membrane, and the blade rests in the depression it makes.
- Surface tension forces exist at liquid/gas and liquid/solid interfaces. They are relatively small, and yet they are decisive in many situations.
- Without surface tension the blade would sink, because its weight is greater than its buoyant force.
- The surface tension forces are not large enough to support a slightly heavier single-edged blade.
- Adding a drop of detergent lowers the surface tension, and the blade sinks at once.

Contact angle and wetting



Contact angle.

- Surface tension acts at the interface between two fluids that do not mix, because the molecular attraction differs on either side of it.
- A drop on a flat surface makes a contact angle with it, set by the surface tension: water on plastic about 90° , alcohol lower and spreading more, mercury well past 90° and spreading less.
- The same wetting shows in a meniscus: water is drawn up at the tube wall, while mercury stands higher at the centre.
- The solid matters too: hydrophobic surfaces repel the liquid, hydrophilic ones attract it. Soap lowers the surface tension of water, which is why it wets fabric.
- The clip ends with a soap-driven boat, pushed by the imbalance in surface tension between bow and stern, and a soap film holding a dense gas inside a pipe.

The pressure inside a drop

Cut a spherical drop in half.

The force developed around the edge by the surface tension is $2\pi R\sigma$, and it must be balanced by the pressure difference acting over the circular area πR^2 :

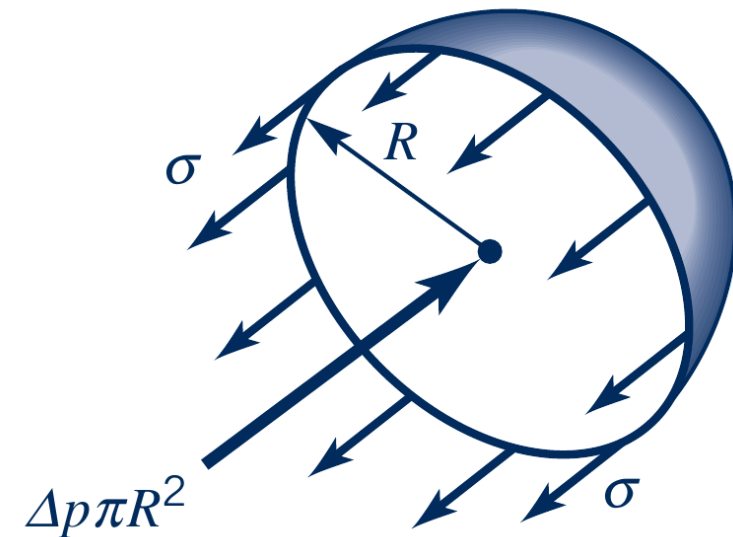
$$2\pi R \sigma = \Delta p \pi R^2$$

$$\Delta p = p_i - p_e = \frac{2\sigma}{R}$$

$$R = 1 \text{ mm} : \quad \Delta p = \frac{2 (0.0728 \text{ N/m})}{1.0 \times 10^{-3} \text{ m}} = 146 \text{ Pa}$$

The pressure inside the drop is higher than outside, and the smaller the drop the larger the difference.

Would the pressure inside a bubble of water be the same as inside a drop of the same diameter?



Forces acting on one half of a liquid drop.

Drop, bubble and cavity

It depends on how many surfaces the liquid has. Each surface pulls with $2\pi R\sigma$ around the cut, and each one has to be balanced by the same pressure difference acting over πR^2 .

➡ DROP

- A ball of liquid in air.
- One surface, the outside of the drop.

$$\Delta p = \frac{2\sigma}{R} = 146 \text{ Pa}$$

➡ BUBBLE

- A thin film of liquid with air inside and outside.
- Two surfaces, inner and outer, so twice the force to balance.

$$\Delta p = \frac{4\sigma}{R} = 291 \text{ Pa}$$

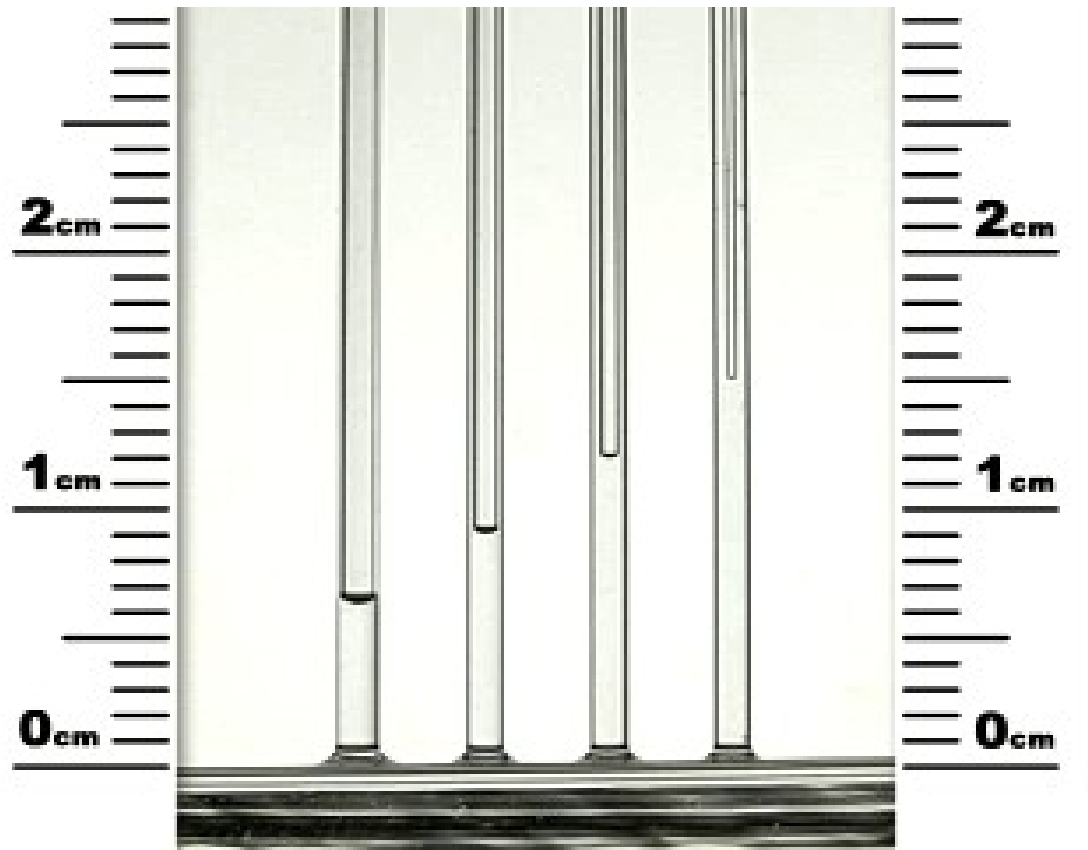
➡ CAVITY

- A gas bubble submerged in a liquid.
- One surface again, so it matches the drop.

$$\Delta p = \frac{2\sigma}{R} = 146 \text{ Pa}$$

So no. A bubble of water of the same diameter needs twice the pressure difference of a drop, because its film has two surfaces. A gas bubble rising in water has only one, and matches the drop exactly. The figures are for water at 20 °C with $R = 1 \text{ mm}$.

Capillary rise



Video © The Regents of the University of California.

- Surface tension is not usually thought of as important in everyday flows, and yet it is critical in many situations — almost all of them on a small length scale.
- The rise of a liquid in a small tube, such as the rise of sap in the narrow channels within the trunk of a tree, is produced by the surface tension forces at the liquid/gas interface inside the tube.
- The height of the capillary rise is inversely proportional to the diameter of the tube.
- The clip shows water rising in glass tubes of several different diameters

Capillary action

Insert a small open tube into water and the level inside rises above the level outside.

The attraction between the wall and the liquid, adhesion, overcomes the mutual attraction of the liquid molecules, cohesion, and the liquid wets the wall.

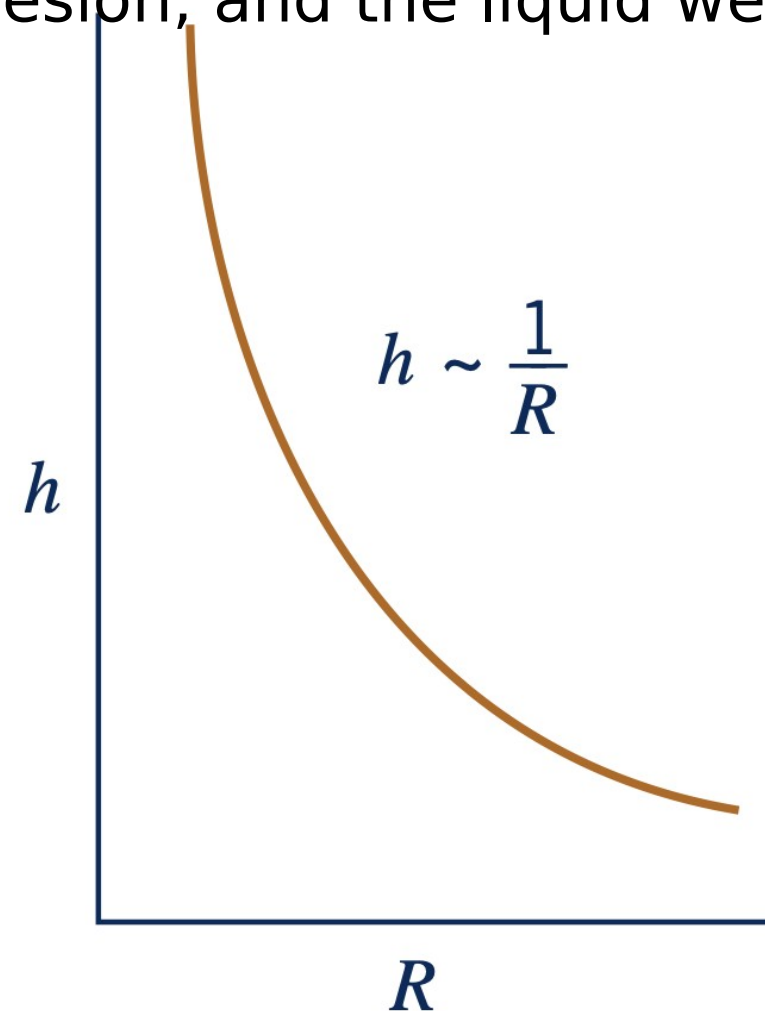
The vertical force from surface tension is $2\pi R\sigma \cos \theta$ and the weight of the raised column is $\gamma\pi R^2h$.

In equilibrium:

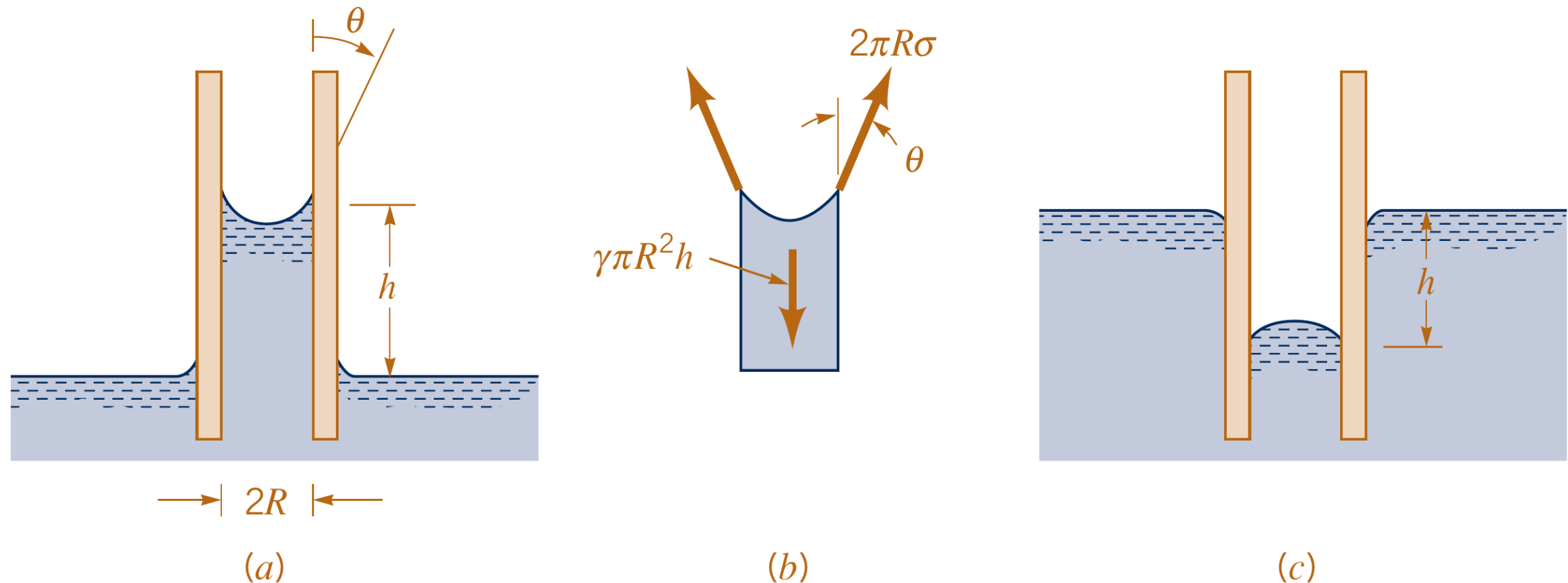
$$\gamma \pi R^2 h = 2\pi R \sigma \cos \theta$$

$$h = \frac{2\sigma \cos \theta}{\gamma R}$$

$$\theta \approx 0^\circ \text{ (water/glass),} \quad \theta \approx 130^\circ \text{ (mercury/glass)}$$



Wetting and non-wetting liquids



(a) Rise of the column for a liquid that wets the tube. (b) Free-body diagram. (c) Depression for a non-wetting liquid.

If adhesion to the solid is weak compared with cohesion, the liquid does not wet the surface and the column is depressed instead.

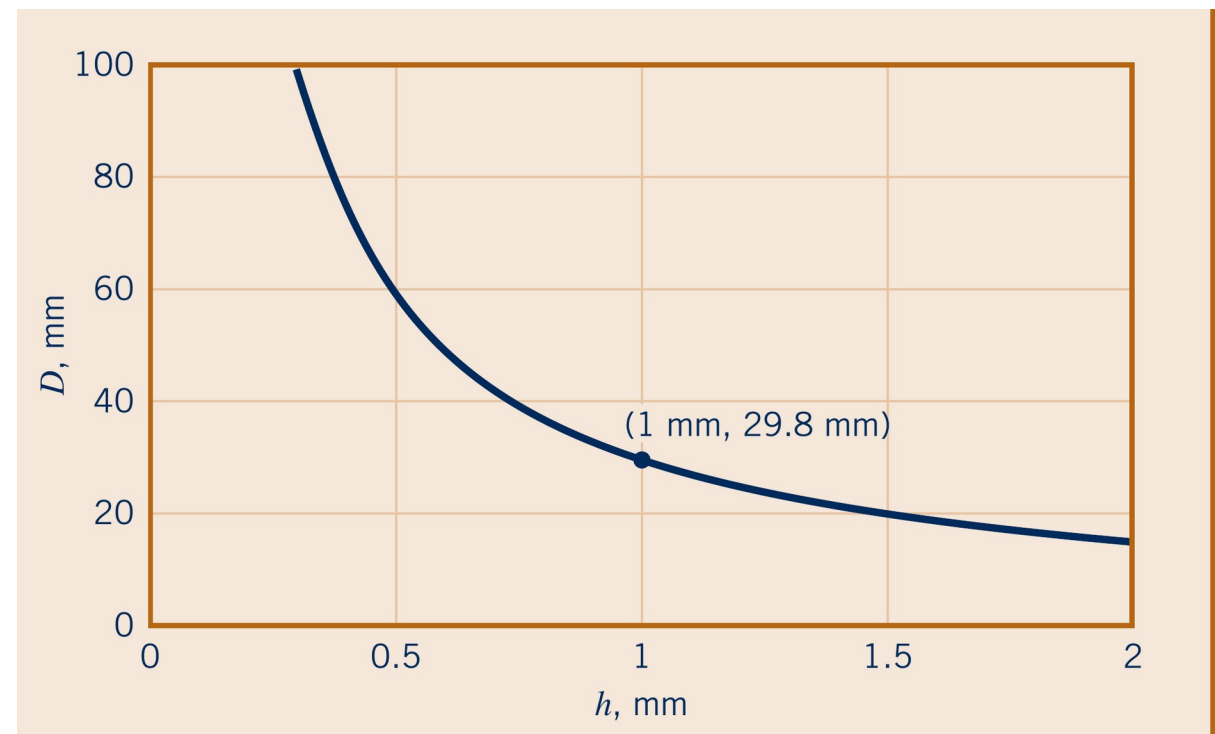
Mercury in a glass tube is the standard example, with a contact angle of about 130° .

Example — capillary rise in a tube

Pressures are often measured with a manometer, by reading the height of a column of liquid in a vertical tube. Determine the diameter of clean glass tubing needed to keep the capillary rise of water at 20 °C below 1.0 mm. Take $\sigma = 0.0728$ N/m, $\gamma = 9.789$ kN/m³ and a contact angle $\theta \approx 0^\circ$.

$$R = \frac{2 (0.0728 \text{ N/m})(1)}{(9.789 \times 10^3 \text{ N/m}^3)(1.0 \times 10^{-3} \text{ m})} = 0.0149 \text{ m}$$

$$D = 2R = 0.0298 \text{ m} = 29.8 \text{ mm}$$



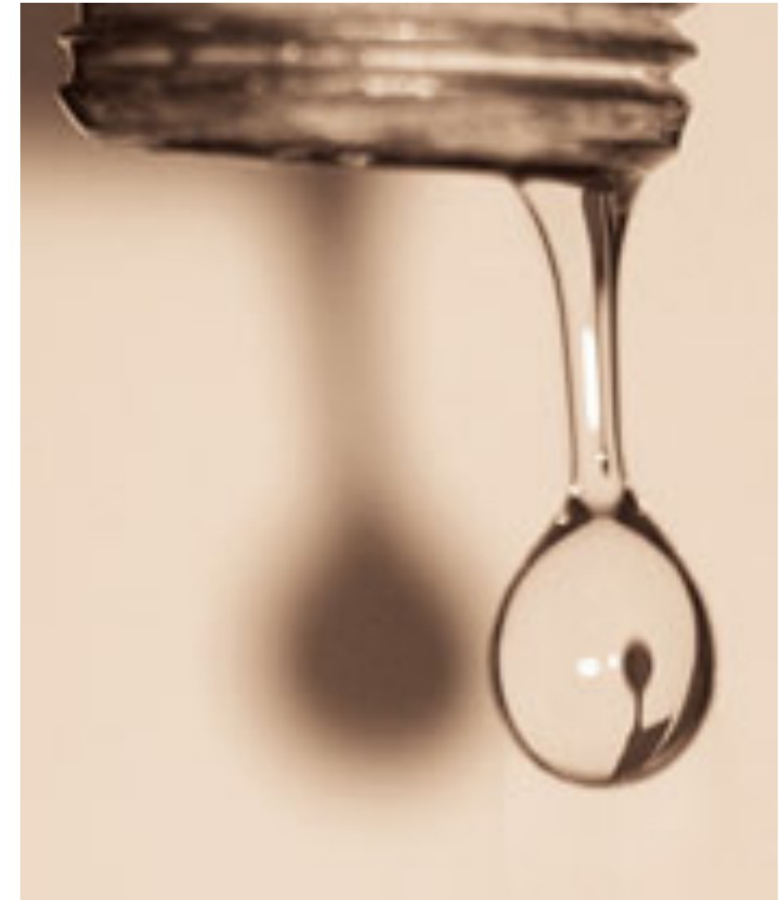
As the allowable rise is reduced, the required diameter grows quickly.

Where surface tension matters

- Movement of liquids through soil and other porous media
- Flow of thin films
- Formation of drops and bubbles
- Breakup of liquid jets
- Drops falling from a leaking faucet

In many problems surface tension is not important: inertia, gravity and viscosity dominate.

Knowing when it can be neglected is part of the modelling.



*Photograph copyright 2007 by Andrew Davidhazy,
Rochester Institute of Technology.*

Ancestral Understanding of Fluid Dynamics

→ ECUADOR

- Albarradas, Santa Elena peninsula: earthen embankments that catch runoff, store it and recharge the aquifer, from Valdivia times onward

bgm.revistas.csic.es/index.php/bgm/article/view/140

- Ingapirca and Pumapungo: Cañari and Inca gravity-fed channels, cisterns and baths

commons.wikimedia.org/wiki/Category:Ingapirca

→ PERU

- Nazca puquios, from about AD 500: spiral shafts and galleries that tap groundwater and deliver it by gravity; several still work

commons.wikimedia.org/wiki/Category:Cantalloc_aqueducts

- Tipón, near Cusco: Inca channels, drops and fountains cut in stone, still running with a steady jet

commons.wikimedia.org/wiki/Category:Tipón



Figura 9. Falla de un tape artesanal durante la época invernal., visto desde diferentes puntos.

Figure 9. Failure of an artisanal “tape” during the winter season, seen from different views.

Credits:

<https://bgm.revistas.csic.es/index.php/bgm/article/view/140/144>

Sailing across the wind: the balsa raft

The Manteño and Huancavilca peoples of the Ecuadorian coast sailed ocean-going balsa rafts. Spanish accounts from 1526 describe them carrying cargo far offshore.

en.wikipedia.org/wiki/Pre-Columbian_rafts

Their control device was the guara: a board lowered between the logs. Raising or lowering the guaras changes how much underwater area there is, and where it sits along the raft.

[en.wikipedia.org/wiki/Guara_\(centerboard\)](https://en.wikipedia.org/wiki/Guara_(centerboard))

- More area forward, and the raft turns away from the wind
- More area aft, and it turns into the wind
- With the guaras set correctly the raft holds a course across the wind, and can work upwind

A hydrodynamic control surface: a side force on a submerged plate, centuries before any theory of lift.



Credits: "Historia Marítima del Ecuador, Tomo II. Periodo prehispánico". (Jenny Estrada, 2012).
<https://acesse.one/s2u218v>

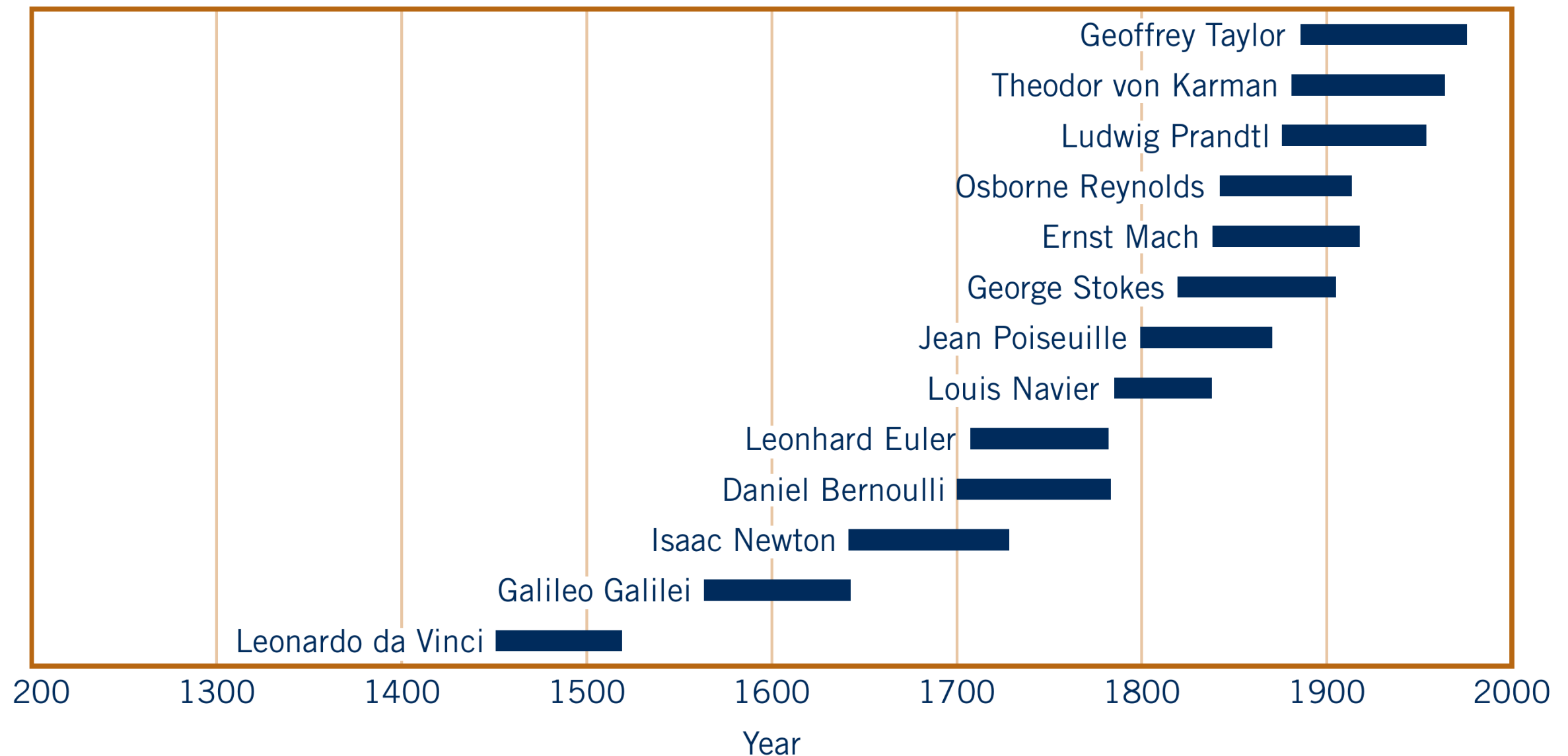
A brief look back in history

- **Archimedes (287–212 B.C.):** the principles of hydrostatics and flotation
- **Sextus Julius Frontinus (A.D. 40–103):** the Roman water supply systems
- **Leonardo da Vinci (1452–1519):** sketches and descriptions of flow phenomena
- **Galileo Galilei (1564–1642):** the beginning of experimental mechanics
- **Newton, Bernoulli, Euler, d'Alembert:** the theoretical foundation

Theory and experiment then developed apart: hydrodynamics for idealized frictionless fluids, hydraulics for the observed behaviour of real water.

In 1904 Ludwig Prandtl (1875–1953) introduced the boundary layer, and the two lines of work were finally joined.

Contributors to the science of fluid mechanics



Time line of some contributors to the science of fluid mechanics.

Every name on this line reappears later in the course, attached to an equation, a dimensionless number or a boundary condition.

Lecture summary

The lecture defined the properties that govern how a fluid responds to pressure and to its own surface: the bulk modulus and the speed of sound, the vapor pressure that sets boiling and cavitation, and the surface tension that lifts a liquid in a narrow tube.

$$E_v = \frac{dp}{d\rho/\rho}$$

bulk modulus

$$E_v = \Gamma p$$

bulk modulus of a gas

$$\frac{p}{\rho^\Gamma} = \text{constant}$$

isentropic process

$$c = \sqrt{\frac{E_v}{\rho}} = \sqrt{\Gamma R T}$$

speed of sound

$$c = \sqrt{\Gamma R T}$$

speed of sound in a gas

$$\text{Ma} = \frac{v}{c}$$

Mach number

$$p = p_v$$

boiling and cavitation

$$\Delta p = \frac{2\sigma}{R}$$

pressure inside a drop

$$h = \frac{2\sigma \cos \theta}{\gamma R}$$

capillary rise